



Drone-borne Retrieval of Cloud Droplet Size Distributions Using Polarimetric Imagery

Introduction

Marine cloud brightening (MCB) is a proposed climate-intervention method that aims to slow global warming by increasing the reflectivity of low-level marine clouds. Small seawater droplets are sprayed into the air. When water evaporates, the small salt crystals generated rise to low-level marine clouds and become additional cloud condensation nuclei, which can make cloud droplets smaller and increase cloud reflectivity through the Twomey effect (Twomey, 1974, 1977).

However, MCB effectiveness remains uncertain because field measurements are limited and many models are too coarse to resolve local cloud microphysical changes. This fourth-year project develops a low-cost drone-borne method for retrieving high-spatial-resolution cloud droplet size distributions (DSDs) in future MCB-relevant field experiments.

Methodology

- Cloud DSDs were modelled using a gamma distribution, described by effective radius r_{eff} and effective variance v_{eff} .
- The scattering-plane polarisation signal in the cloudbow region depends strongly on scattering angle θ and cloud DSD, and is modelled using the Mueller-matrix polarisation phase function $P_{12}(\theta)$.
- The cloudbow region is where $135^\circ \leq \theta \leq 165^\circ$.
- A calibrated polarisation camera measures a proxy for $P_{12}(\theta)$. $\theta = 180^\circ - \gamma$, where γ is the viewing-angle difference between each cloud pixel and the detected anti-solar point (ASP).
- Two retrieval modes were tested: 'single-frame retrieval' uses data from different cloud patches in one frame, and 'multi-frame retrieval' tracks the same cloud patch across frames.
- Measured profiles were fitted to theoretical $P_{12}(\theta)$ lookup-table (LUT) profiles generated for different r_{eff} and v_{eff} , following Pörtge et al. (2023).
- See Figure 1 to the right for more details.

Drone Platform and Fieldwork

- LUCID 5 MP polarisation camera mounted on an FPV drone.
- Raspberry Pi 4B records 12-bit video at 10 FPS to SSD storage.
- Camera payload powered separately from the drone.
- Fieldwork carried out above valley fog in the Yorkshire Dales National Park, UK.

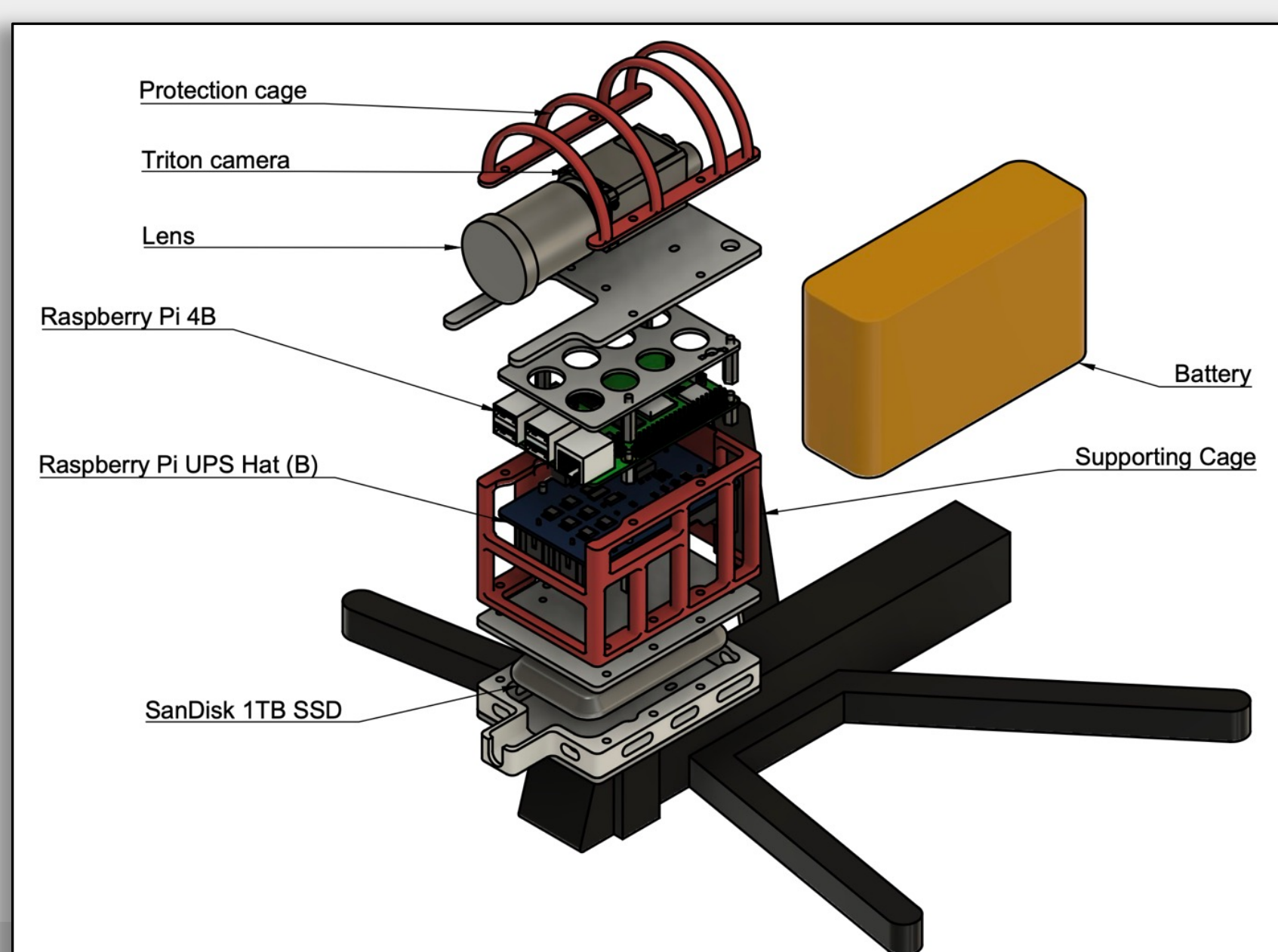


Figure 2. Exploded view of the drone used for retrieval.

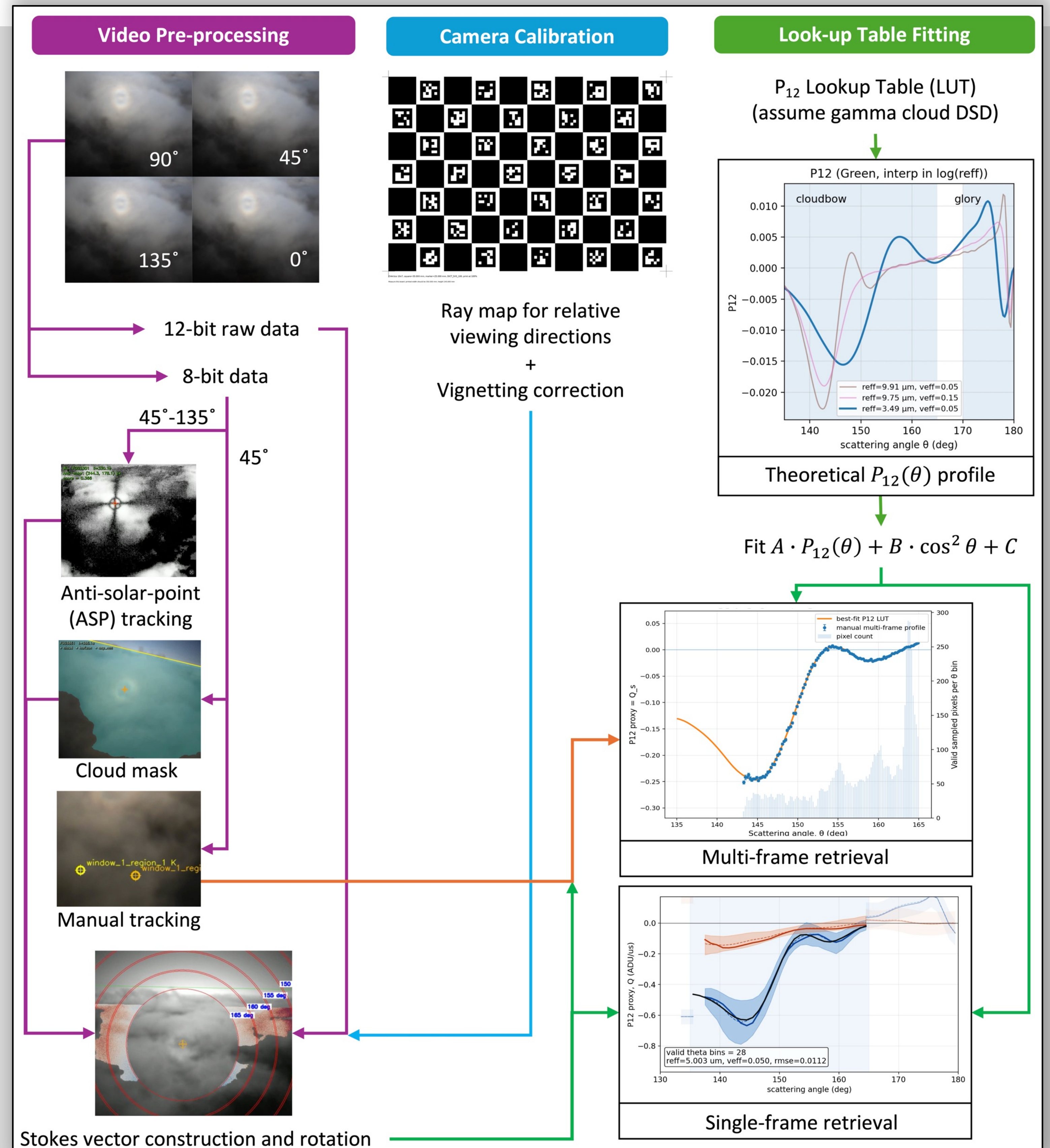


Figure 1. Flowchart for the retrieval of cloud DSD using lookup table and recorded polarimetric videos.

Results and Discussion

- Single-frame and multi-frame retrievals gave consistent DSDs, with $r_{eff} \approx 5 \mu\text{m}$ and $v_{eff} \approx 0.05$. The retrieved r_{eff} is close to reported monomodal valley-fog values of $5.5 \mu\text{m}$ (Mazoyer et al., 2019; Mazoyer, Burnet and Denjean, 2022).
- The multi-frame retrieval achieved approximately $3 \text{ m} \times 3 \text{ m}$ spatial resolution, around one to two orders of magnitude finer than current satellite or airborne polarimetric cloud DSD retrieval methods.
- The multi-frame LUT fit quality is high, with RMSE approximately 1% of the peak signal amplitude.
- This low-cost method could support future MCB fieldwork, satellite cloud-product validation, and fog or low-cloud model evaluation.

Limitations and Future Work

- Automate cloud-patch tracking to make multi-frame retrieval scalable.
- Improve radiometric calibration and ensure in-focus image acquisition.
- Improve field of view using a wider lens, or calibrate drone attitude/IMU data to mosaic frames when the ASP is not visible.
- Add cloud-motion correction and more robust cloud masking.



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